



EE3220 SYSTEM-ON-CHIP (SOC) DESIGN

TEAM-BASED LEARNING (TBL) RISC-V MISSION CLOCK IMPLEMENTATION

23 MARCH 2026

PURPOSE OF THIS TBL

- Apply SoC design concepts
- Integrate RISC-V core, UART, timer, firmware
- Practice MMIO and system integration
- Build a working system under time pressure

SCHEDULE AND FLOW

- iRAT: 3:00 – 3:15pm
- Team Based Project: 3:15 – 5:30pm
- Target Completion: 5:30pm
- Presentation: 5:30 – 5:45pm
- Award Ceremony: 5:45 – 5:50pm

SYSTEM ARCHITECTURE

- Given:
 - RISC-V CPU, Memory
- Implement:
 - MMIO UART
 - MMIO Timer
 - Firmware

WHAT SUCCESS LOOKS LIKE

- Start at 00:00:00
- Update every second
- Correct rollover
- Optional: VT100/ANSI
- Optional: Clock setting via UART / Pushbuttons

GRADING AND EXPECTATION

- Functional correctness mandatory
- Creative designs receives bonuses
- FPGA demo
- System integration
- TAs will NOT debug your design